

Inflammation and Autoimmunity

Cytokines and inflammation, HLA, systemic autoimmunity, intestine, skin and joints, thyroid and diabetes type 1, granulomas, lung and innate immunity

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. Polygenic scores	5
4. Variants by theme	7
5. HLA inferred (statistical)	11
6. Favourable	12
7. Unfavourable	13
8. Mechanisms of ageing	14
9. Practical suggestions	15
10. Limitations	16
11. Methods and sources	17
12. Annex: all variants analysed	18

1. Summary

What was seen. 70 SNPs were read in 56 genes (all of the chip) and 29 polygenic scores. The general picture is of low to typical autoimmune profile: HLA is unusually "clean" and no autoimmune disease has a clearly increased genetic risk, with two slight exceptions (sarcoidosis and vitiligo).

Favourable. Without markers of **B*27**, **C*06:02**, **DQ2.5**, **DQ8**, **DRB1*15:01** and **DRB1*04** (shared epitope). **PTPN22** and **STAT4** without risk alleles (protection of almost all autoimmunes). **NOD2** without R702W or G908R; **MUC5B** without pulmonary fibrosis allele (OR ~5); **TNF -308 G/G**, **NLRP3** without Q705K, **IL6R Ala358** (less IL-6 signaling) and 1 allele of low CRP. Low scores: IBD p1.7 and p9.3, celiac p3.9, psoriasis p6.3 and p2.4, Sjögren p3.2, AR p18.6, IL-8 p17, 4.

Unfavourable. CARD9 risk homozygous in 2 SNPs (IBD, spondylitis, IgA nephropathy; small effect). Sarcoidosis p92.3, consistent with **BTNL2** and homozygous risk **ANXA11**. Vitiligo p89, 9. Uveite p84.5 (without B*27). Gastric pathway more reactive: **PSCA T/T**, **IL1B -511T**, **CXCL8 -251 A/A**, TLR1/TLR10. Innate immunity: **SERPINA1 Pi*Z** heterozygous (probable Pi*MZ), MBL2 codon 52 heterozygous, **IFIH1 T/T**. Two HLA SNPs of homozygous ulcerative colitis, compensated by the rest (ulcerous colitis p27.3).

Reading. An adaptive immune system with little autoimmune predisposition and an innate/granulomatous immunity a little more reactive. The points to be monitored are the skin (vitiligo), eyes (uveitis), lungs and ganglia (sarcoidosis) and stomach (if there is H. pylori).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (statistical imputation, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 70 SNPs: 70 of the chip, 0 imputed and 0 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 5 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizygous).
- **Polygenic scores (PGS):** 29 PGS Catalog scores calculated locally with 2,780,678 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$), calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

Percentages in the reference population. The scores with low coverage (50% celiac, LES 37%) are marked as low reliability. The lupus and celiac depend heavily on HLA, which the chip reads only in part.

Inflammatory markers and cells

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
CRP (score A)	PGS000314 (Circ Genom Precision Med 2020)	76 77 (weighted coverage 99%)	25.9	TYPICAL	typical
CRP (score B)	PGS002873 (Am J Hum Genet 2022)	1,082,460 1,113,831 (weighted coverage 96%)	25.8	TYPICAL	typical
IL-6	PGS000252 (Nat Metab 2020)	541,930 552,426 (weighted coverage 95%)	46.0	TYPICAL	typical
IL-18	PGS000249 (Nat Metab 2020)	541,739 552,186 (weighted coverage 98%)	65.4	TYPICAL	typical
IL-8	PGS000254 (Nat Metab 2020)	541,477 551,990 (weighted coverage 98%)	17.4	LESS LIKELY	favourable
Neutrophils	PGS000182 (Cell 2020)	388 492 (weighted coverage 80%)	20.5	TYPICAL	without judgment
Leukocytes	PGS000191 (Cell 2020)	513 636 (weighted coverage 82%)	64.1	TYPICAL	without judgment

Intestine and celiac

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Inflammatory bowel disease (score A)	PGS004105 (Am J Hum Genet 2024)	137 139 (weighted coverage 98%)	1.7	LESS LIKELY	favourable
Inflammatory bowel disease (B score)	PGS001288 (PLoS Genet 2022)	156 193 (weighted coverage 85%)	9.3	LESS LIKELY	favourable
Crohn's disease	PGS001331 (PLoS Genet 2022)	211 256 (weighted coverage 84%)	48.3	TYPICAL	typical
Ulcerative colitis	PGS001306 (PLoS Genet 2022)	141 176 (weighted coverage 82%)	27.3	TYPICAL	typical
Celiac disease	PGS000040 (PLoS Genet 2014)	122 228 (weighted coverage 50%)	3.9	LESS LIKELY	favourable (low reliability)

Skin and joints

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Rheumatoid arthritis (score A)	PGS004118 (Am J Hum Genet 2024)	89 91 (weighted coverage 97%)	39.2	TYPICAL	typical
Rheumatoid arthritis (score B)	PGS001875 (Am J Hum Genet 2021)	249 256 (weighted coverage 99%)	18.6	LESS LIKELY	favourable
Psoriasis (score A)	PGS002293 (J Am Acad Dermal 2022)	61 62 (weighted coverage 99%)	6.3	LESS LIKELY	favourable
Psoriasis (score B)	PGS002083 (Am J Hum Genet 2021)	70,315 71,744 (weighted coverage 99%)	2.4	LESS LIKELY	favourable
Psoriatic arthritis	PGS000198 (Sci Transl Med 2020)	17 18 (weighted coverage 96%)	40.4	TYPICAL	typical
Ankylosing spondylitis (score A)	PGS001876 (Am J Hum Genet 2021)	81 85 (weighted coverage 75%)	73.1	TYPICAL	typical
Ankylosing spondylitis (score B)	PGS003421 (J Transl Med 2022)	11 16 (weighted coverage 65%)	5.1	LESS LIKELY	favourable
Vitiligo	PGS000738 (Am J Hum Genet 2019)	46 48 (weighted coverage 90%)	89.9	MORE LIKELY	unfavourable

Systemic autoimmunity and others

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Systemic lupus erythematosus	PGS000328 (Ann Rheum Dis 2020)	24 57 (weighted coverage 37%)	78.7	TYPICAL	typical (low reliability)
Systemic sclerosis	PGS000341 (Ann Rheum Dis 2021)	33 33 (weighted coverage 100%)	47.2	TYPICAL	typical
Multiple sclerosis	PGS000809 (EJHG 2021)	101 127 (weighted coverage 84%)	23.9	TYPICAL	typical
1 type diabetes	PGS000021 (Diabetes Care 2016)	25 25 (weighted coverage 100%)	41.9	TYPICAL	typical
Sarcoidosis	PGS001872 (Am J Hum Genet 2021)	74 79 (weighted coverage 93%)	92.3	MORE LIKELY	unfavourable
Sjögren syndrome	PGS001308 (PLoS Genet 2022)	7 7 (weighted coverage 100%)	3.2	LESS LIKELY	favourable
Uveitis	PGS019061 (Nature 2025)	18 20 (weighted coverage 93%)	84.5	MORE LIKELY	unfavourable

Thyroid

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Lymphocytic thyroiditis	PGS005270 (medRxiv 2025)	50 55 (weighted coverage 70%)	33.8	TYPICAL	typical
Hypothyroidism	PGS000759 (Nat Common 2021)	127 140 (weighted coverage 82%)	41.4	TYPICAL	typical

4. Variants by theme

For each SNP: genotype (strand +), origin, effect allele and number of copies, judgment and evidence. In the "protective" SNPs, having the allele is favourable.

Cytokines and inflammation

Genes of cytokines that fix the inflammatory tone: TNF, IL-1 β , IL-6, IL-10, IL-17, IL-8, NLRP3, CRP and NF- κ B (TNFAIP3/A20).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TNF	rs1800629	-308G>A	G/G	chip	A (0 copy/s)	favourable	Allele The high producer of TNF- α	High
LTA	rs909253	+252	A/A	chip	G (0 copy/s)	favourable	High G allele TNF/LT- α	Low
IL1B	rs16944	-511 (-511T in gene)	A/G	chip	A (1 copy/s)	unfavourable (light)	Proinflammatory allele: plus IL-1 β and more gastritis with H. pylori	Moderate
IL1B	rs1143634	+3954	G/G	chip	A (0 copy/s)	favourable	High producer allele IL-1 β	Moderate
IL6	rs1800795	-174	G/G	chip	G (2 copy/s)	unfavourable	Old association to higher IL-6; inconsistent	Low
IL6R	rs2228145	Asp358Ala	A/C	chip	C (1 copy/s)	favourable	Ala358: less IL-6 signaling in the membrane (coronary disease protection and AR)	High
IL10	rs1800896	-1082 (G in the gene, high producer)	T/C	chip	C (1 copy/s)	neutral	Allele of higher IL-10 production (anti-inflammatory)	Moderate
IL10/IL19	rs3024505	intergenic	G/A	chip	A (1 copy/s)	unfavourable (light)	Less IL-10; risk of ulcerative colitis	High
IL17A	rs2275913	-197	G/G	chip	A (0 copy/s)	favourable	High IL-17A producer allele	Moderate
CXCL8 (IL-8)	rs4073	-251	A/A	chip	A (2 copy/s)	unfavourable	Higher production of IL-8 (neutrophil attraction)	Low
NLRP3	rs35829419	Q705K	C/C	chip	A (0 copy/s)	favourable	NLRP3 inflammasome function gain	High
CRP	rs1205	3'UTR	C/T	chip	T (1 copy/s)	favourable	T allele: genetically lower CRP	High
CRP	rs1130864	3'UTR	G/G	chip	A (0 copy/s)	favourable	Allele up to CRP	Moderate
TNFAIP3	rs6920220	6q23	G/A	chip	A (1 copy/s)	unfavourable (light)	Less brake of NF- κ B (A20); risk of AR and LES	High

HLA (classical allele markers)

HLA is the strongest genetic factor of autoimmunity. Here, markers (tags) of classical alleles are used; they do not replace a HLA typing.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HLA-B*27 (tag)	rs4349859	tag B*27	G/G	chip	A (0 copy/s)	favourable	Ankylosing spondylitis and acute anterior uveitis	High
HLA-B*27 (tag)	rs116488202	tag B*27	C/C	chip	T (0 copy/s)	favourable	Ankylosing spondylitis	High
HLA-C*06:02 (tag)	rs10484554	tag C*06:02	C/C	chip	T (0 copy/s)	favourable	Early onset psoriasis	High

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HLA-DQA1*05:01 (DQ2.5)	rs2187668	tag DQ2.5/DR3	C/C	chip	T (0 copy/s)	favourable	Celiac disease, 1 type diabetes, membranous nephropathy	High
HLA-DQ8 (tag)	rs7454108	tag DQ8	T/T	chip	C (0 copy/s)	favourable	Celiac disease and 1 type diabetes	High
HLA-DRB1*15:01 (tag)	rs3135388	tag DRB1*15:01	G/G	chip	A (0 copy/s)	favourable	Multiple sclerosis and optic neuritis	High
HLA-DRB1*04 (tag)	rs660895	shared epitope	A/A	chip	G (0 copy/s)	favourable	Seropositive rheumatoid arthritis	High
HLA-DQA1/DRA	rs6927022	class II	A/A	chip	A (2 copy/s)	unfavourable	Ulcerative colitis (OR ~1.4 by allele)	High
HLA-DRA/DQA2	rs2395185	class II	G/G	chip	G (2 copy/s)	unfavourable	Ulcerative colitis	Moderate

Systemic autoimmunity (LES, AR, Sjögren)

Genes shared by several systemic autoimmune diseases: lymphocyte signaling (PTPN22, CTLA4, SH2B3, BLK), interferon pathway (STAT4, IRF5, TYK2) and immunocomplex clearance (ITGAM, NCF2).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PTPN22	rs2476601	R620W	G/G	chip	A (0 copy/s)	favourable	Risk of almost all autoimmune (AR, LES, DT1, thyroid, vitiligo)	High
STAT4	rs7574865	intronic	G/G	chip	T (0 copy/s)	favourable	LES, AR, Sjögren and systemic sclerosis	High
IRF5	rs2004640	splice	G/G	chip	T (0 copy/s)	favourable	LES and Sjögren	High
IRF5	rs4728142	promoter	G/A	chip	A (1 copy/s)	unfavourable (light)	LES, Sjögren and ulcerative colitis	High
ITGAM	rs1143679	R77H	G/A	chip	A (1 copy/s)	unfavourable (light)	LES (OR ~1.7)	High
NCF2	rs17849502	H389Q	G/T	chip	T (1 copy/s)	unfavourable (light)	LES (rare variant; OR ~1.5–2)	Moderate
BLK	rs13277113	promoter	G/A	chip	A (1 copy/s)	unfavourable (light)	LES and systemic sclerosis	High
TNFSF4 (OX40L)	rs2205960	intergenic	G/T	chip	T (1 copy/s)	unfavourable (light)	LES and systemic sclerosis	High
TRAF1-C5	rs3761847	intergenic	G/G	chip	G (2 copy/s)	unfavourable	Rheumatoid arthritis (OR ~1.1–1.3)	High
CTLA4	rs231775	+49 (Ala17)	A/A	chip	G (0 copy/s)	favourable	Autoimmune thyroid, DT1 and AR	High
CTLA4	rs3087243	CT60	G/A	chip	G (1 copy/s)	unfavourable (light)	AR, DT1 and autoimmune thyroid	High
SH2B3	rs3184504	R262W	C/C	chip	T (0 copy/s)	favourable	Celiac, DT1 and autoimmune thyroid	High
TYK2	rs34536443	P1104A	G/G	chip	C (0 copy/s)	neutral	Psoriasis protective allele, LES and AR (partial loss of function)	High

Bowel (inflammatory bowel disease and celiac)

Autophagy and bacterial sensors of the intestine (NOD2, ATG16L1, IRGM, CARD9), IL-23 axis and celiac loci outside the HLA.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
NOD2	rs2066844	R702W	C/C	chip	T (0 copy/s)	favourable	Main Crohn risk variant	High
NOD2	rs2066845	G908R	G/G	chip	C (0 copy/s)	favourable	Crohn's risk variable	High
IL23R	rs11209026	R381Q	G/G	chip	A (0 copy/s)	neutral	IBD protective allele, spondylitis and psoriasis	High
ATG16L1	rs2241880	Thr300Ala	A/G	chip	G (1 copy/s)	unfavourable (light)	Autophagy: Crohn (OR ~1.3)	High
CARD9	rs4077515	S12N	T/T	chip	T (2 copy/s)	unfavourable	Antifungal and neutrophil signaling: IBD, spondylitis, nephropathy IgA (OR ~1.2/ allele)	High
CARD9	rs10781499	intronic	A/A	chip	A (2 copy/s)	unfavourable	IBD, spondylitis and nephropathy IgA	High
IRGM	rs13361189	promoter	T/T	chip	C (0 copy/s)	favourable	Autophagy: Crohn	High
PTGER4	rs4613763	5p13	T/T	chip	C (0 copy/s)	favourable	Crohn and multiple sclerosis	High
JAK2	rs10758669	intronic	C/A	chip	C (1 copy/s)	unfavourable (light)	IBD	High
MST1	rs3197999	R689C	G/A	chip	A (1 copy/s)	unfavourable (light)	IBD and sclerosing cholangitis	High
IL2-IL21	rs6822844	4q27	G/T	chip	T (1 copy/s)	favourable	Protective T allele of celiac, AR and DT1	High

Skin and joints (psoriasis, spondylitis)

Presentation of antigens (ERAP1) and IL-17 signaling in the skin (TRAF3IP2, TNIP1, LCE3).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ERAP1	rs30187	K528R	C/C	chip	T (0 copy/s)	favourable	Allele of higher activity of ERAP1: spondylitis and psoriasis (with B27/C06)	High
TRAF3IP2	rs33980500	D10N	C/C	chip	T (0 copy/s)	favourable	Psoriasis and psoriatic arthritis	High
TNIP1	rs17728338	5q33	G/G	chip	A (0 copy/s)	favourable	Psoriasis, psoriatic arthritis and LES	High
LCE3B/C (deletion)	rs4112788	deletion tag	A/A	chip	G (0 copy/s)	favourable	Psoriasis (LCE3C_LCE3B)	Moderate

Thyroid, 1 type diabetes and multiple sclerosis

Loci of 1 type diabetes, multiple sclerosis and autoimmune thyroid disease outside HLA.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
IFIH1	rs1990760	A946T	T/T	chip	T (2 copy/s)	unfavourable	Antiviral response of the strongest interferon type DT1 (OR ~1.15)	High
PTPN2	rs1893217	intronic	A/A	chip	G (0 copy/s)	favourable	DT1 and IBD	High
IL2RA	rs2104286	intronic	T/T	chip	T (2 copy/s)	unfavourable	DT1 and multiple sclerosis (common allele)	High
CLEC16A	rs12708716	intronic	A/G	chip	A (1 copy/s)	unfavourable (light)	DT1 and multiple sclerosis	High
UBASH3A	rs11203203	intronic	G/A	chip	A (1 copy/s)	unfavourable (light)	DT1	High
IL7R	rs6897932	T244I	C/C	chip	C (2 copy/s)	unfavourable	Multiple sclerosis (common allele; OR ~1.1)	High
TNFRSF1A	rs1800693	splice	T/T	chip	C (0 copy/s)	favourable	Multiple sclerosis	High

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TSHR	rs179247	intronic	A/G	chip	A (1 copy/s)	unfavourable (light)	Graves' disease	High
FOXE1	rs965513	9q22	A/G	chip	A (1 copy/s)	unfavourable (light)	Hypothyroidism and thyroid cancer	High

Granulomas, lung, stomach and innate immunity

Granulomatous diseases (sarcoidosis), pulmonary fibrosis, gastric inflammatory response and innate immunity (MBL2, alpha-1-antitrypsin).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BTNL2	rs2076530	splice (A in gene)	T/T	chip	T (2 copy/s)	unfavourable	Sarcoidosis (OR ~1.6/allele)	Moderate
ANXA11	rs1049550	R230C (Arg230)	G/G	chip	G (2 copy/s)	unfavourable	Sarcoidosis (OR ~1.3/allele)	Moderate
MUC5B	rs35705950	promoter	G/G	chip	T (0 copy/s)	favourable	Idiopathic pulmonary fibrosis (OR ~5)	High
DSP	rs2076295	intronic	T/T	chip	G (0 copy/s)	favourable	Pulmonary fibrosis	High
SERPINA1	rs28929474	Pi*Z (Glu342Lys)	C/T	chip	T (1 copy/s)	unfavourable (light)	Less inhibition of neutrophil elastase (probable Pi*MZ)	High
MBL2	rs5030737	codon 52 (D)	G/A	chip	A (1 copy/s)	unfavourable (light)	Lower mannose binding leukemia (infections)	High
MBL2	rs1800450	codon 54 (B)	C/C	chip	T (0 copy/s)	favourable	Lowest mannose binding lectin	High
PSCA	rs2294008	promoter	T/T	chip	T (2 copy/s)	unfavourable	Atrophic gastritis and diffuse gastric cancer (OR ~1.3/allele); protects from duodenal ulcer	High
ABO	rs505922	The probable group	T/T	chip	T (2 copy/s)	without judgment	Group O: more duodenal ulcer with H. pylori; less thromboembolism	Moderate
TLR1/TLR10	rs10004195	4p14	T/A	chip	A (1 copy/s)	unfavourable (light)	Higher seroprevalence of H. pylori	Moderate

5. HLA inferred (statistical)

In addition to the markers SNPs, the previous analysis inferred the classic HLA alleles from the neighboring chip SNPs, with cross-validation in a reference panel. It is a statistical inference, not a typing.

Result	Alleles
Probable carrier	DRB1*11, DRB1*13, DQB1*03:01, A*29
Possible non-carrier	B*27, C*06:02, DRB1*15:01, DQB1*06:02, DRB1*03:01, DQB1*02 (DQ2), DQB1*03:02 (DQ8), DRB1*04, DRB1*01:01, DRB1*10:01, DRB1*07:01, DRB1*08, B*08, A*01, B*51, B*57, B*38, B*39, C*12, DQB1*05, A*02

Interpretation: DRB1*11 is added to BTNL2 and ANXA11 in the mild risk of sarcoidosis. A*29 is present in almost all birdshot chorioretinopathy patients, a very rare disease (very low absolute risk). An HLA typing of 2 fields would confirm these alleles and also the pharmacogenetic ones (B*58:01, A*31:01).

6. Favourable

- **Clean HLA:** without markers of B*27 (2 tags), C*06:02, DQ2.5/DR3, DQ8, DRB1*15:01 and DRB1*04.
- **Systemic autoimmunity:** PTPN22 without R620W, STAT4 and IRF5 rs2004640 without risk, CTLA4 +49 A/A, SH2B3 R262 homozygous.
- **Intestine:** NOD2 without R702W or G908R; IRGM and PTGER4 without risk; IL2-IL21 with 1 protective allele. IBD p1.7 and p9.3; celiac p3.9 (low reliability).
- **Skin and joints:** ERAP1 Lys528 homozygous (low activity), TRAF3IP2 and TNIP1 without risk, LCE3B/C probably without deletion. Psoriasis p6.3 and p2.4; spondylitis p5.1 (score B); AR p18.6 (score B).
- **Cytokines:** TNF –308 G/G, LTA +252 A/A, IL1B +3954 G/G, IL17A without the high producer allele, NLRP3 without Q705K, **IL6R Ala358** heterozygous, **CRP rs1205 T** (CRP lower); IL-8 p17.4; CRP p25.9.
- **Thyroid and DT1/EM:** PTPN2 and TNFRSF1A without risk; EM p23.9; hypothyroidism p41.4.
- **Lung:** MUC5B G/G and DSP without risk (pulmonary fibrosis).
- **Sjögren p3.2.**

7. Unfavourable

- **CARD9** rs4077515 T/T and rs10781499 A/A: more active antifungal and neutrophil signaling (IBD, spondylitis, IgA nephropathy; OR ~1.2/allele). Compensated by low IBD scores.
- **Sarcoidosis p92.3**, with **BTNL2 T/T** and **ANXA11 G/G** (Arg230) homozygous risk and probable DRB1*11. Low absolute risk (rare disease).
- **Vitiligo p89.9** and uveitis p84.5 (without B*27).
- **Gastric route: PSCA T/T** (atrophic gastritis), **IL1B –511T** (1), **CXCL8 –251 A/A**, TLR1/TLR10 (1) and O group: major gastric inflammatory response if there is H. pylori.
- **Innate immunity: SERPINA1 Pi*Z** (1 copy; less brake on elastase), MBL2 codon 52 (1); MBL intermediate), **IFIH1 A946T T/T** (stronger antiviral response; DT1).
- **HLA of ulcerative colitis:** rs6927022 A/A and rs2395185 G/G, with IL10 rs3024505 (1); ulcerative colitis p27.3.
- **LES:** ITGAM R77H, NCF2 H389Q, IRF5 rs4728142, BLK and TNFSF4 (1 copy each); LES p78.7 (low reliability).
- **Minors:** TRAF1-C5 G/G and TNFAIP3 (1) for AR ATG16L1 Thr300Ala, JAK2, MST1 (1 each); IL2RA T/T and IL7R C/C (common alleles of DT1/EM); CLEC16A, UBASH3A, TSHR, FOXE1 and CTLA4 CT60 (1 each); IL6 –174 G/G (weak).

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Chronic Inflammation	Calm-based cytokines (TNF, IL-17A, NLRP3, IL6R Ala358, CRP rs1205 T); IL1B -511T and CXCL8 A/A in the opposite direction	favourable
Inflammaging	PCR p25,9, IL-6 p46,0, IL-8 p17,4, IL-18 p65.4	favourable
Immunosenescence	IFIH1 T/T (more active interferon), MBL2 heterozygous codon 52	unfavourable (light)
Intercellular Communication	PTPN22, STAT4, CTLA4 and SH2B3 without risk; HLA without the main autoimmune alleles	favourable
Intestinal Barrier dysfunction	NOD2/IRGM without risk; CARD9 homozygous risk; IBD p1.7	mixed
Autophagy / Deregulated Macroautophagy	ATG16L1 Thr300Ala (1 copy); IRGM without risk	unfavourable (light)
Remodelling of the extracellular matrix	SERPINA1Pi*Z heterozygous: less inhibition of elastase (lung, liver)	unfavourable
Fragmentation of Elastin	Probable Pi*MZ: neutrophil elastase has less brake	unfavourable
Dysbiosis	Reactive gastric route with H. pylori (PSCA, IL1B, CXCL8, TLR1); FUT2 secretor	unfavourable (light)
Neuroinflammation	EM p23.9; DRB1*15:01 absent; TNFRSF1A without risk	favourable
Thyroid alterations	Thiroiditis p33.8 and hypothyroidism p41, 4 TSHR and FOXE1 with 1 allele	neutral

9. Practical suggestions

All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Research of H. pylori** (breathing test or antigen in feces), once, and eradication if positive: PSCA T/T, IL1B -511T and CXCL8 A/A make chronic gastritis more likely if the bacterium is present. Suggestion (not applied).
2. **Confirm Pi*MZ** with serum alpha-1-antitrypsin and the Pi phenotype; never smoke or vaporize and avoid dust and smoke. Suggestion (not applied).
3. **HLA typing of 2 fields**, once: confirms DRB1*11, A*29 and alleles of pharmacogenetics. Suggestion (not applied).
4. **Directed surveillance, without extra routine exams:** Depigmented spots (vitiligo), painful red eye or blurred vision (uveitis), persistent dry cough or increased ganglia (sarcoidosis). Any of these signs justifies consultation. Suggestion (not applied).
5. **Anti-inflammatory lifestyle** (already in the plan): a lot of fiber, rich in vegetables, legumes, fish and olive oil, omega-3 3.2-4 g/day of EPA+DHA, exercise, sleep and zero alcohol. Suggestion (not applied).
6. **Vitamin D sufficient** (already in the stack), with modulating effect on autoimmune diseases. Suggestion (not applied).
7. **Vaccination** according to the plan (pneumococcal conjugate, flu, COVID, recombinant zoster, hepatitis B): Heterozygous MBL2 and IFIH1 T/T. Suggestion (not applied).
8. **Rapamycin already present in the stack** is immunomodulatory; these genes do not change their use, but repeated infections should be reported to the physician. Suggestion (not applied).

10. Limitations

- Small and added effects; a high score changes little the absolute risk of rare diseases (sarcoidosis, uveitis).
- HLA is read only by SNP markers and by statistical inference; it is not a typing.
- The celiac and lupus scores have low coverage (50% and 37%).
- The NOD2 1007fs probe (insertion) was not used; a previous reading suggested absence, confirming if there is clinical suspicion.
- JAK2 V617F and clonal hematopoiesis are somatic and do not read on the chip.
- Conclusions only from genes; do not include blood analysis.

11. Methods and sources

- **Genes and SNPs:** 70 SNPs chosen from the 181 SNPs cured from the previous analysis (1 314 lines in total), giving priority to functional variants and loci with high evidence; topics treated in other reports (histamine, pigmentation, neurological diseases) were removed.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report. No genetic data came out of the analysis computer.
- **Guidance:** Alleles of effect in strand +, conferred against the alleles of the site; 5 palindromic SNPs assumed in the strand +.
- **Polygenic scores:** PGS Catalog, calculated with the imputed genome (statistical imputation, public panel of complete genomes to 30x); percentiles in the reference population. Some percentiles differ slightly from previous analysis (e.g. celiac p3.9 instead of p12,5, sarcoidosis p92.3 instead of p94), because it uses the imputed genome instead of inference by nearby neighbors.
- **Comparison with previous analysis (02-10-2026):** 70 genotypes are the same.
- **Context Literature:** GWAS Catalog, genetic revisions of autoimmune diseases (e.g. Cotsapas 2011, Jostins 2012, Fingerlin 2014 for sarcoidosis).

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Cytokines and inflammation	TNF	rs1800629	G/G	chip	A	0	strand +	without the risk allele
Cytokines and inflammation	LTA	rs909253	A/A	chip	G	0	strand +	without the risk allele
Cytokines and inflammation	IL1B	rs16944	A/G	chip	A	1	strand +	1 copy of risk allele
Cytokines and inflammation	IL1B	rs1143634	G/G	chip	A	0	strand +	without the risk allele
Cytokines and inflammation	IL6	rs1800795	G/G	chip	G	2	strand + (palindromic; assumed +)	2 risk allele copies
Cytokines and inflammation	IL6R	rs2228145	A/C	chip	C	1	strand +	1 copy of the protective allele
Cytokines and inflammation	IL10	rs1800896	T/C	chip	C	1	strand +	heterozygous: intermediate IL-10 producer
Cytokines and inflammation	IL10/IL19	rs3024505	G/A	chip	A	1	strand +	1 copy of risk allele
Cytokines and inflammation	IL17A	rs2275913	G/G	chip	A	0	strand +	without the risk allele
Cytokines and inflammation	CXCL8 (IL-8)	rs4073	A/A	chip	A	2	strand + (palindromic; assumed +)	2 risk allele copies
Cytokines and inflammation	NLRP3	rs35829419	C/C	chip	A	0	strand +	without the risk allele
Cytokines and inflammation	CRP	rs1205	C/T	chip	T	1	strand +	1 copy of the protective allele
Cytokines and inflammation	CRP	rs1130864	G/G	chip	A	0	strand +	without the risk allele
Cytokines and inflammation	TNFAIP3	rs6920220	G/A	chip	A	1	strand +	1 copy of risk allele
HLA (classical allele markers)	HLA-B*27 (tag)	rs4349859	G/G	chip	A	0	strand +	without the risk allele
HLA (classical allele markers)	HLA-B*27 (tag)	rs116488202	C/C	chip	T	0	strand +	without the risk allele
HLA (classical allele markers)	HLA-C*06:02 (tag)	rs10484554	C/C	chip	T	0	strand +	without the risk allele
HLA (classical allele markers)	HLA-DQA1*05:01 (DQ2.5)	rs2187668	C/C	chip	T	0	strand +	without the risk allele
HLA (classical allele markers)	HLA-DQ8 (tag)	rs7454108	T/T	chip	C	0	strand +	without the risk allele
HLA (classical allele markers)	HLA-DRB1*15:01 (tag)	rs3135388	G/G	chip	A	0	strand +	without the risk allele
HLA (classical allele markers)	HLA-DRB1*04 (tag)	rs660895	A/A	chip	G	0	strand +	without the risk allele
HLA (classical allele markers)	HLA-DQA1/DRA	rs6927022	A/A	chip	A	2	strand +	2 risk allele copies
HLA (classical allele markers)	HLA-DRA/DQA2	rs2395185	G/G	chip	G	2	strand +	2 risk allele copies
Systemic autoimmunity (LES, AR, Sjögren)	PTPN22	rs2476601	G/G	chip	A	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Systemic autoimmunity (LES, AR, Sjögren)	STAT4	rs7574865	G/G	chip	T	0	strand +	without the risk allele
Systemic autoimmunity (LES, AR, Sjögren)	IRF5	rs2004640	G/G	chip	T	0	strand +	without the risk allele
Systemic autoimmunity (LES, AR, Sjögren)	IRF5	rs4728142	G/A	chip	A	1	strand +	1 copy of risk allele
Systemic autoimmunity (LES, AR, Sjögren)	ITGAM	rs1143679	G/A	chip	A	1	strand +	1 copy of risk allele
Systemic autoimmunity (LES, AR, Sjögren)	NCF2	rs17849502	G/T	chip	T	1	strand +	1 copy of risk allele
Systemic autoimmunity (LES, AR, Sjögren)	BLK	rs13277113	G/A	chip	A	1	strand +	1 copy of risk allele
Systemic autoimmunity (LES, AR, Sjögren)	TNFSF4 (OX40L)	rs2205960	G/T	chip	T	1	strand +	1 copy of risk allele
Systemic autoimmunity (LES, AR, Sjögren)	TRAF1-C5	rs3761847	G/G	chip	G	2	strand +	2 risk allele copies
Systemic autoimmunity (LES, AR, Sjögren)	CTLA4	rs231775	A/A	chip	G	0	strand +	without the risk allele
Systemic autoimmunity (LES, AR, Sjögren)	CTLA4	rs3087243	G/A	chip	G	1	strand +	1 copy of risk allele
Systemic autoimmunity (LES, AR, Sjögren)	SH2B3	rs3184504	C/C	chip	T	0	strand +	without the risk allele
Systemic autoimmunity (LES, AR, Sjögren)	TYK2	rs34536443	G/G	chip	C	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
Bowel (inflammatory bowel disease and celiac)	NOD2	rs2066844	C/C	chip	T	0	strand +	without the risk allele
Bowel (inflammatory bowel disease and celiac)	NOD2	rs2066845	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
Bowel (inflammatory bowel disease and celiac)	IL23R	rs11209026	G/G	chip	A	0	strand +	without the protective allele (common genotype)
Bowel (inflammatory bowel disease and celiac)	ATG16L1	rs2241880	A/G	chip	G	1	strand +	1 copy of risk allele
Bowel (inflammatory bowel disease and celiac)	CARD9	rs4077515	T/T	chip	T	2	strand +	2 risk allele copies
Bowel (inflammatory bowel disease and celiac)	CARD9	rs10781499	A/A	chip	A	2	strand +	2 risk allele copies

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Bowel (inflammatory bowel disease and celiac)	IRGM	rs13361189	T/T	chip	C	0	strand +	without the risk allele
Bowel (inflammatory bowel disease and celiac)	PTGER4	rs4613763	T/T	chip	C	0	strand +	without the risk allele
Bowel (inflammatory bowel disease and celiac)	JAK2	rs10758669	C/A	chip	C	1	strand +	1 copy of risk allele
Bowel (inflammatory bowel disease and celiac)	MST1	rs3197999	G/A	chip	A	1	strand +	1 copy of risk allele
Bowel (inflammatory bowel disease and celiac)	IL2-IL21	rs6822844	G/T	chip	T	1	strand +	1 copy of the protective allele
Skin and joints (psoriasis, spondylitis)	ERAP1	rs30187	C/C	chip	T	0	strand +	without the risk allele
Skin and joints (psoriasis, spondylitis)	TRAF3IP2	rs33980500	C/C	chip	T	0	strand +	without the risk allele
Skin and joints (psoriasis, spondylitis)	TNIP1	rs17728338	G/G	chip	A	0	strand +	without the risk allele
Skin and joints (psoriasis, spondylitis)	LCE3B/C (deletion)	rs4112788	A/A	chip	G	0	strand +	without the risk allele
Thyroid, 1 type diabetes and multiple sclerosis	IFIH1	rs1990760	T/T	chip	T	2	strand +	2 risk allele copies
Thyroid, 1 type diabetes and multiple sclerosis	PTPN2	rs1893217	A/A	chip	G	0	strand +	without the risk allele
Thyroid, 1 type diabetes and multiple sclerosis	IL2RA	rs2104286	T/T	chip	T	2	strand +	2 risk allele copies
Thyroid, 1 type diabetes and multiple sclerosis	CLEC16A	rs12708716	A/G	chip	A	1	strand +	1 copy of risk allele
Thyroid, 1 type diabetes and multiple sclerosis	UBASH3A	rs11203203	G/A	chip	A	1	strand +	1 copy of risk allele
Thyroid, 1 type diabetes and multiple sclerosis	IL7R	rs6897932	C/C	chip	C	2	strand +	2 risk allele copies
Thyroid, 1 type diabetes and multiple sclerosis	TNFRSF1A	rs1800693	T/T	chip	C	0	strand +	without the risk allele
Thyroid, 1 type diabetes and multiple sclerosis	TSHR	rs179247	A/G	chip	A	1	strand +	1 copy of risk allele
Thyroid, 1 type diabetes and multiple sclerosis	FOXE1	rs965513	A/G	chip	A	1	strand +	1 copy of risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Granulomas, lung, stomach and innate immunity	BTNL2	rs2076530	T/T	chip	T	2	strand +	2 risk allele copies
Granulomas, lung, stomach and innate immunity	ANXA11	rs1049550	G/G	chip	G	2	strand +	2 risk allele copies
Granulomas, lung, stomach and innate immunity	MUC5B	rs35705950	G/G	chip	T	0	strand +	without the risk allele
Granulomas, lung, stomach and innate immunity	DSP	rs2076295	T/T	chip	G	0	strand +	without the risk allele
Granulomas, lung, stomach and innate immunity	SERPINA1	rs28929474	C/T	chip	T	1	strand +	1 copy of risk allele
Granulomas, lung, stomach and innate immunity	MBL2	rs5030737	G/A	chip	A	1	strand +	1 copy of risk allele
Granulomas, lung, stomach and innate immunity	MBL2	rs1800450	C/C	chip	T	0	strand +	without the risk allele
Granulomas, lung, stomach and innate immunity	PSCA	rs2294008	T/T	chip	T	2	strand +	2 risk allele copies
Granulomas, lung, stomach and innate immunity	ABO	rs505922	T/T	chip	T	2	strand +	2 copy of the effect allele
Granulomas, lung, stomach and innate immunity	TLR1/TLR10	rs10004195	T/A	chip	A	1	strand + (palindromic; assumed +)	1 copy of risk allele